geometry is still some significant steps away from the blank-slate state anticipated above, but it does give us a hint of what the more complete framework for incorporating space and time may involve.

Finding the correct mathematical apparatus for formulating string theory without recourse to a pre-existing notion of space and time is one of the most important issues facing string theorists. An understanding of how space and time emerge would take us a huge step closer to answering the crucial question of which geometrical form actually does emerge.

Will String Theory Lead to a Reformulation of Quantum Mechanics?

The universe is governed by the principles of quantum mechanics to fantastic accuracy. Even so, in formulating theories over the past half century, physicists have followed a strategy that, structurally speaking, places quantum mechanics in a somewhat secondary position. In devising theories, physicists often start by working in a purely classical language that ignores quantum probabilities, wave functions, and so forth—a language that would be perfectly intelligible to physicists in the age of Maxwell and even in the age of Newton—and then, subsequently, overlaying quantum concepts upon the classical framework. This approach is not particularly surprising, since it directly mirrors our experiences. At first blush, the universe appears to be governed by laws rooted in classical concepts such as a particle having a definite position and a definite velocity at any given moment in time. It is only after detailed microscopic scrutiny that we realize that we must modify such familiar classical ideas. Our process of discovery has gone from a classical framework to one that is modified by quantum revelations, and this progression is echoed in the way that physicists, to this day, go about constructing their theories.

This is certainly the case with string theory. The mathematical formalism describing string theory begins with equations that describe the motion of a tiny, infinitely thin piece of classical thread—equations that, to a large extent, Newton could have written down some three hundred years ago. These equations are then quantized. That is, in a systematic manner developed by physicists over the course of more than 50 years, the classical equations are converted into a quantum-mechanical framework in which probabilities, uncertainty, quantum